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# Paleosecular variation of brunhes age lava flows from British Columbia, Canada

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[1] Brunhes age lava flows have been sampled for paleosecular variation studies from several volcanic fields of southern British Columbia (Silverthrone, Garibaldi Lake, Mt. Meager, Clearwater-Wells Gray Park and Kelowna area). A total of 52 lava flows were sampled and 7 to 10 samples were drilled at each site. Previous radiometric studies indicate that the ages of these lava flows range from 2.3 to 760 Ka. Stepwise thermal demagnetization (14 to 21 steps) was carried out for all the samples in each site and AF demagnetization was performed on one sample per site. Forty-five sites were selected based on rigorous criteria ($\alpha_{95} \leq 5°$) to calculate a mean direction (D = 356.9°, I = 70.2°, $\alpha_{95}$ = 2.8°) that is statistically indistinguishable from the direction of the dipole field at the area (I = 68.3°). Virtual geomagnetic poles (VGPs) do not show the far-sided effect and the angular standard deviation is 17.5°, a value in agreement with the paleomagnetic field dispersion for that latitude. These high quality results are expected to improve the time averaged field (TAF) and secular variation models.

**Components:** 6438 words, 7 figures, 2 tables.

**Keywords:** Virtual geomagnetic pole; secular variation; geocentric axial dipole; Garibaldi Volcanic Belt; lava flow.

**Index Terms:** 1522 Geomagnetism and Paleomagnetism: Paleomagnetic secular variation; 1560 Geomagnetism and Paleomagnetism: Time variations—secular and long term; 1530 Geomagnetism and Paleomagnetism: Rapid time variations.

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**Theme:** Geomagnetic Field Behavior Over the Past 5 Myr
**Guest Editors:** Cathy Constable and Catherine Johnson

## 1. Introduction

[2] This study is part of a larger study referred to as the time averaged dipole field initiative (TAFI) which intends to provide a series of high quality paleomagnetic studies on lava flows up to 5 Ma, along the backbone of the Americas, from the Arctic to the Antarctic. This initiative emerged

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upon realizing the need to improve the paleomagnetic data set to better constrain time-averaged field (TAF) and secular variation models. TAF models have suggested the presence of nonzonal structures in the paleomagnetic field that persist over long periods of time [e.g., *Johnson and Constable*, 1995, 1997] but their nature is not well defined because the data sets on which they are based are incomplete or inadequate. Likewise secular variation models in which the dispersion of the virtual geomagnetic pole increases with latitude need to be refined [*Johnson and Constable*, 1996] using better data.

[3] TAF and secular variation models have been based on paleomagnetic data sets [e.g., *McElhinny and McFadden*, 1997; *Johnson and Constable*, 1996] that have been compiled applying certain selection criteria. For example paleomagnetic studies from individual areas in which blanket or no demagnetization have been applied are not included. The selection criteria for acceptance of paleomagnetic results are somewhat arbitrary, although necessary. With increasing attention placed on achieving very accurate results, recent paleomagnetic results [e.g., *Carlut et al.*, 2000; *Johnson et al.*, 1998; *Tauxe et al.*, 2000] have greatly superseded previous selection criteria. *Tauxe et al.* [2000] set up new selection criteria in which the $\alpha_{95}$ of individual sites is restricted to 5° (for comparison *McElhinny and McFadden* [1997] used 20°). Our results fulfill *Tauxe et al.*'s [2000] quality criteria for individual sites. It is therefore expected that such selection criteria will produce data sensitive to nondipole (5–10% of the Earth's field) components of the field.

[4] Here, paleomagnetic results from 53 sites from southern British Columbia (latitude 50°–51.5°N) are presented. Samples were taken from the Silverthrone, Mount Meager and Garibaldi Lake volcanic fields of the Garibaldi Volcanic Belt, as well as from the Wells Gray-Clearwater volcanic fields of the Anahim Volcanic Belt and the Kelowna area field of the Chilcotin Plateau Basalts (Figure 1). These volcanic areas rest on metamorphic rocks and were produced in a variety of tectonic settings: arc volcanism (Garibaldi Volcanic Belt); hot spot volcanism (Anahim Volcanic Belt) and back arc volcanism (Chilcotin Plateau Basalts). Volcanism in this area both predates and postdates Pleistocene glaciations. During the glacial and interglacial periods, sediments of glacial, fluvial and lacustrine environments were deposited along with volcanic products, including the lava flows that are the focus of this study. A rich variety of geomorphic shapes such as tuya volcanoes, horns, glacial valleys, and moraines are a legacy of the glacial age. Many of the lava flows studied here are valley-filling basalts and display a wide variety of sub-aqueous and sub aerial textures. Age control was achieved by (1) paleomagnetic sampling of previously studied lava flows and (2) accepting the age of the volcanic unit given in the literature. Based on these ages, the sampled lava flows represent mostly the last 550 ka although flows ranging from 550 to 760 ka, were also sampled.

[5] The commonly used scatter of the virtual geomagnetic poles (VGPs) relative to the Earth's axis of rotation is calculated for the studied lava flows as an expression of secular variation. The result is compared with that expected according to Model G of secular variation [*McFadden et al.*, 1988]. Time averaged fields are calculated and analyzed based on the selection criteria and other spatial and stratigraphic relations among the sites. The presence of the "far sided effect" [*Wilson*, 1970] attributed to a axial quadrupole harmonic of the field [*McElhinny et al.*, 1996a] is evaluated, although the effect of negative inclination anomalies expected to be contributed by an axial quadrupole harmonic, superposed on a predominantly dipolar field, is relatively small at the latitude of the area studied (e.g., inclination anomaly is −1° for an axial quadrupole amounting 5% of the GAD). Likewise the results are compared with existing time averaged field (TAF) moldels and with results from Brunhes age lavas of the Indian Heaven volcanic field (Washington State) obtained by *Mitchell et al.* [1989], which are of high quality.

## 2. Sampling and Geologic Description

[6] A total of 53 paleomagnetic sites were sampled by Barendregt during the summers of 1998 and 1999. Samples were taken using a hand held drill and oriented using magnetic compass and, when possible, sun compass (30% of the samples). From

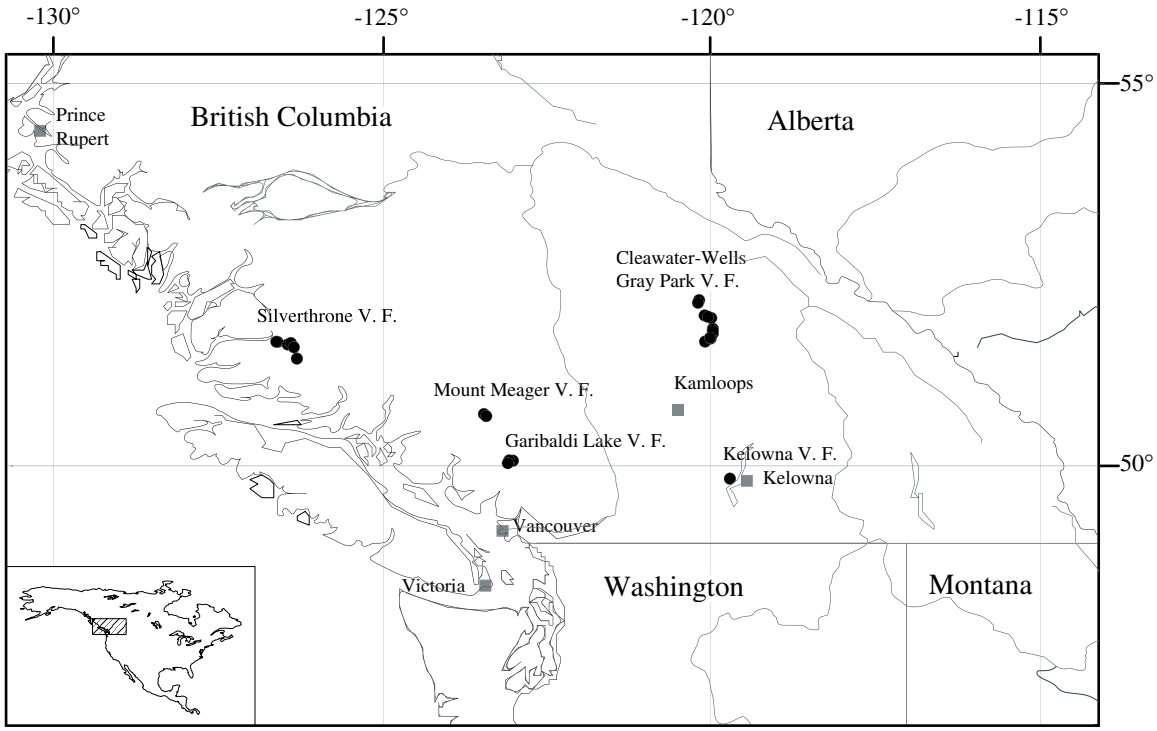



**Figure 1.** Location map showing sampling sites (black dots) and their correspondent volcanic fields.

8 to 10 cores were collected from all sites, each site corresponding to a single lava flow. Site location was determined using GPS. Most sites were accessed by road and short hikes were necessary. A helicopter was used to access two flows. Most of the sampled lava flows fill valleys and are exposed as a result of subsequent erosion. Other lava flows were sampled closer to volcanoes and are at a higher altitude. Several sites form series of overlapping lava flows, generally from 3 to 9 in number. The lava sequences and their stratigraphic relationship are given in Table 1.

[7] The Silverthrone volcanic field (Figure 2) was studied by *Green et al.* [1988] who defined three units: (1) breccias overlying metamorphic rocks in angular discontinuity, (2) ryolitic, dacitic and andesitic flows overlying the breccia, and (3) unconsolidated fluvial and volcano-sedimentary deposits resting on deeply eroded flow surfaces. The main focus of sampling in this volcanic field were the basaltic andesite flows which originated from numerous centers and filled the Machmell and Pashleth river valleys and are locally interstratified with sediments (unit 3). Sites 16-1 and 16-2 were sampled from older andesitic flows near Kingcome Glacier (unit 2). An age of 0.4 Ma (K-Ar) was obtained by *Green et al.* [1988] at a nearby site. The remaining 15 paleomagnetic sites from this volcanic field were obtained from the valley filling basaltic andesitic flows of unit 3. The radiometric dates obtained for some of these basaltic flows place them in the last 150 Ka [*Green et al.*, 1988]. Site 5-1 was taken from a lava flow associated with a boulder covered in part by barnacles which yielded a 12 Ka $^{14}C$ radiometric date [*Roddick*, 1996]. Sites from this volcanic field are indicated in Table 1 by the abbreviation ST.

[8] The Garibaldi Lake Volcanic Field was studied by *Green et al.* [1988] and comprises nine volcanic centers that have been active through the Quaternary. Three flows were sampled from the volcanic center that occupies the Cheakamus river valley. One of these flows (site 1-3) corresponds to a unit that locally contains pebbles striated in two directions suggesting that the unit predates the Fraser Glaciation ice sheet, which has an age of 50 Ka. The other two sites (sites 1-4 and 1-5) correspond to lava flows that overlie the previously described

**Table 1.** Paleomagnetic and Age Data From Sites[a]

| Site ID | U/L | DG | V.F. | Site Lat. | Site Long. | D | I | SC | n/N | K Dir. | $\alpha_{95}$ Dir. | VGP Lat. | VGP Long. | $\alpha_{95}$ VGP | K VGP | AF/Th | Age, ka | U (+/−), ka | R | Meth. |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1-1 | | | MM | 50.68 | −123.48 | 23.2 | 53.2 | 10 | 6/10 | 642 | 2.6 | 66.1 | 2.5 | 3.3 | 410 | Th | 2.34 | 0.05 | 1 | $^{14}$C |
| 1-2* | | | MM | 50.65 | −123.44 | 23.1 | 56.2 | 10 | 10/10 | 110 | 4.6 | 68.8 | 357.1 | 5.7 | 73 | Th | 2.34 | 0.05 | 1 | $^{14}$C |
| 1-3 | | | GL | 50.07 | −123.04 | 41.3 | 77.5 | 12 | 11/12 | 350 | 2.4 | 63.6 | 273.7 | 4.4 | 109 | Th | 50 to 150 | | 2 | ST |
| 1-4 | U | | GL | 50.07 | −123.09 | 342.1 | 68.7 | 10 | 10/10 | 401 | 2.4 | 78.5 | 164.1 | 3.9 | 157 | Th | <34.2 | | 3 | $^{14}$C |
| 1-5 | L | | GL | 50.04 | −123.11 | 323.1 | 77.0 | 10 | 10/10 | 502 | 2.2 | 65.6 | 199.5 | 3.9 | 153 | Th | <34.2 | | 3 | $^{14}$C |
| 2-1 | U | | ST | 51.62 | −126.62 | 343.4 | 68.1 | 0 | 6/7 | 146 | 5.6 | 79.8 | 148.4 | 8.8 | 60 | Th | ≤150 | | 2 | K/Ar |
| 2-2 | | | ST | 51.62 | −126.62 | 46.0 | 56.4 | 0 | 7/7 | 327 | 3.3 | 54.7 | 329.3 | 3.9 | 241 | Th | ≤150 | | 2 | K/Ar |
| 2-3 | L | | ST | 51.62 | −126.62 | 311.5 | 74.0 | 0 | 6/7 | 21 | 14.8 | 62.7 | 180.2 | 26.2 | 7 | Th | ≤150 | | 2 | K/Ar |
| 5-1 | | | ST | 51.62 | −126.60 | 358.5 | 67.3 | 0 | 5/7 | 488 | 3.5 | 88.3 | 87.2 | 5.0 | 234 | Th | 12 | | 4 | $^{14}$C |
| 7-1 | | | ST | 51.59 | −126.45 | 6.3 | 61.3 | 0 | 7/7 | 68 | 7.4 | 80.7 | 22.2 | 9.5 | 41 | Th | ≤150 | | 2 | K/Ar |
| 8-1 | | | ST | 51.59 | −126.45 | 5.1 | 56.1 | 0 | 7/7 | 523 | 2.6 | 74.7 | 37.7 | 3.4 | 322 | Th | ≤150 | | 2 | K/Ar |
| 9-1 | L | a | ST | 51.61 | −126.40 | 1.6 | 67.9 | 0 | 7/7 | 696 | 2.3 | 88.8 | 356.7 | 3.5 | 301 | Th | ≤150 | | 2 | K/Ar |
| 9-2 | | – | ST | 51.61 | −126.40 | 356.2 | 72.8 | 0 | 6/6 | 102 | 6.7 | 82.3 | 217.5 | 11.5 | 35 | Th | ≤150 | | 2 | K/Ar |
| 9-3 | | – | ST | 51.61 | −126.40 | 357.9 | 63.6 | 0 | 7/7 | 93 | 6.3 | 83.6 | 65.2 | 9.7 | 40 | Th | ≤150 | | 2 | K/Ar |
| 9-4 | | a | ST | 51.61 | −126.40 | 8.9 | 72.2 | 0 | 6/7 | 480 | 3.1 | 82.2 | 272.1 | 5.0 | 180 | Th | ≤150 | | 2 | K/Ar |
| 9-5 | U | | ST | 51.61 | −126.40 | 1.3 | 68.3 | 0 | 6/7 | 409 | 3.3 | 89.2 | 326.9 | 5.5 | 152 | Th | ≤150 | | 2 | K/Ar |
| 10-1 | | | ST | 51.55 | −126.35 | 12.6 | 68.9 | 0 | 8/8 | 206 | 3.9 | 82.1 | 311.8 | 6.2 | 80 | Th | ≤150 | | 2 | K/Ar |
| 13-1 | | | ST | 51.59 | −126.43 | 23.1 | 67.9 | 0 | 5/7 | 352 | 4.1 | 75.6 | 316.9 | 6.6 | 137 | Th | ≤150 | | 2 | K/Ar |
| 14-1 | | | ST | 51.59 | −126.43 | 13.0 | 68.0 | 0 | 7/8 | 254 | 3.8 | 82.0 | 320.7 | 5.9 | 107 | Th | ≤150 | | 2 | K/Ar |
| 15-1 | | | ST | 51.59 | −126.43 | 14.7 | 68.7 | 0 | 7/7 | 391 | 3.1 | 80.9 | 313.6 | 4.7 | 177 | Th | ≤150 | | 2 | K/Ar |
| 16-1 | U | | ST | 51.40 | −126.31 | 348.3 | 58.3 | 0 | 5/10 | 476 | 3.5 | 75.2 | 91.7 | 4.4 | 310 | Th | 400 | | 2 | K/Ar |
| 16-2 | L | | ST | 51.40 | −126.31 | 357.1 | 51.3 | 0 | 10/10 | 96 | 4.9 | 70.7 | 60.2 | 6.0 | 65 | Th | 400 | | 2 | K/Ar |
| 18-1 | | | CW | 52.13 | −120.23 | 5.7 | 69.2 | 1 | 8/8 | 151 | 4.5 | 86.4 | 309.3 | 7.3 | 59 | AF | 300 | 300 | 5 | K/Ar |
| 19-1 | | | CW | 52.17 | −120.22 | 23.6 | 71.6 | 0 | 6/10 | 576 | 2.8 | 75.6 | 303.3 | 4.4 | 231 | AF | 300 | 300 | 5 | K/Ar |
| 20-1 | U | | CW | 51.97 | −120.13 | 4.0 | 64.3 | 0 | 9/11 | 172 | 3.9 | 83.6 | 32.1 | 6.0 | 75 | Th | 200/280 | 110/150 | 6 | K/Ar |
| 20-2 | L | | CW | 51.97 | −120.13 | 25.5 | 63.0 | 0 | 10/12 | 152 | 3.9 | 71.8 | 343.6 | 5.8 | 70 | Th | 200/280 | 110/150 | 6 | K/Ar |
| 21-1 | U | b | CW | 51.95 | −120.08 | 15.4 | 85.8 | 0 | 9/12 | 898 | 1.7 | 60.0 | 244.4 | 3.4 | 232 | Th | 200/280 | 110/150 | 6 | K/Ar |
| 21-2 | L | b | CW | 51.95 | −120.08 | 78.9 | 88.0 | 0 | 8/11 | 155 | 4.5 | 52.5 | 246.2 | 8.8 | 40 | Th | 200/280 | 110/150 | 6 | K/Ar |
| 22-1 | U | c | CW | 51.93 | −120.03 | 26.2 | 73.5 | 0 | 7/10 | 610 | 2.4 | 73.6 | 292.3 | 4.2 | 203 | Th | 561 | 106 | 6 | K/Ar |
| 22-2 | | c | CW | 51.93 | −120.03 | 20.4 | 76.2 | 11 | 9/11 | 459 | 2.4 | 73.9 | 273.7 | 4.3 | 145 | Th | 561 | 106 | 6 | K/Ar |
| 22-3 | L | c | CW | 51.93 | −120.03 | 9.7 | 74.6 | 10 | 8/10 | 157 | 4.4 | 79.1 | 265.2 | 7.5 | 55 | Th | 561 | 106 | 6 | K/Ar |
| 23-1 | | | CW | 51.79 | −120.01 | 312.0 | 66.4 | 9 | 3/12 | 291 | 7.2 | 59.9 | 164.0 | 10.3 | 145 | Th | 547 | 102 | 6 | K/Ar |

**Table 1.** (continued)

| Site ID | U/L | DG | V.F. | Site Lat. | Site Long. | D | I | SC | n/N | K Dir. | $\alpha_{95}$ Dir. | VGP Lat. | VGP Long. | $\alpha_{95}$ VGP | K VGP | AF/Th | Age, ka | U (+/−), ka | R | Meth. |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 24-1 | | | CW | 51.77 | −120.01 | 330.8 | 62.0 | 0 | 4/8 | 166 | 7.1 | 69.0 | 138.4 | 9.4 | 96 | AF | 547 | 102 | 6 | K/Ar |
| 25-1 | | | CW | 51.73 | −120.01 | 348.3 | 66.7 | 0 | 11/12 | 323 | 2.5 | 82.1 | 138.1 | 3.9 | 140 | th | 547 | 102 | 6 | K/Ar |
| 25-2 | U | | CW | 51.73 | −120.01 | 336.3 | 69.3 | 0 | 10/10 | 596 | 2 | 75.6 | 164.5 | 3.2 | 222 | th | 547 | 102 | 6 | K/Ar |
| 25-3 | | | CW | 51.73 | −120.01 | 329.7 | 64.0 | 0 | 9/10 | 665 | 2 | 69.3 | 145.3 | 2.8 | 329 | th | 547 | 102 | 6 | K/Ar |
| 25-4 | | | CW | 51.73 | −120.01 | 359.6 | 68.4 | 0 | 10/10 | 290 | 2.8 | 89.6 | 141.8 | 4.6 | 112 | th | 547 | 102 | 6 | K/Ar |
| 25-5 | | | CW | 51.73 | −120.01 | 341.9 | 70.0 | 0 | 8/10 | 814 | 1.9 | 78.8 | 168.9 | 3.0 | 338 | th | 547 | 102 | 6 | K/Ar |
| 25-6 | | d | CW | 51.73 | −120.01 | 327.6 | 70.9 | 8 | 10/10 | 398 | 2.4 | 70.6 | 173.7 | 4.0 | 148 | th | 547 | 102 | 6 | K/Ar |
| 25-7 | | d | CW | 51.73 | −120.01 | 324.8 | 66.9 | 0 | 10/10 | 239 | 3.1 | 67.8 | 158.6 | 4.9 | 98 | th | 547 | 102 | 6 | K/Ar |
| 25-8 | | e | CW | 51.73 | −120.01 | 351.9 | 66.5 | 3 | 7/10 | 220 | 4.1 | 84.4 | 126.9 | 6.0 | 102 | th | 547 | 102 | 6 | K/Ar |
| 25-9 | | e | CW | 51.73 | −120.01 | 350.3 | 66.5 | 0 | 10/10 | 237 | 3.1 | 83.3 | 133.5 | 4.4 | 122 | th | 547 | 102 | 6 | K/Ar |
| 25-10 | L | e | CW | 51.73 | −120.01 | 350.6 | 65.1 | 0 | 11/11 | 134 | 4 | 82.8 | 120.3 | 5.7 | 65 | th | 547 | 102 | 6 | K/Ar |
| 26-1 | L | f | CW | 51.68 | −120.05 | 350.5 | 68.7 | 10 | 8/10 | 401 | 2.8 | 84.1 | 157.6 | 4.5 | 152 | th | 350 | 90 | 7 | K/Ar |
| 26-2 | | f | CW | 51.68 | −120.05 | 344.1 | 66.7 | 10 | 7/10 | 226 | 4 | 79.8 | 143.5 | 6.3 | 93 | th | 350 | 90 | 7 | K/Ar |
| 26-3 | U | f | CW | 51.68 | −120.05 | 351.8 | 61.8 | 10 | 5/10 | 333 | 4.2 | 79.9 | 96.4 | 5.1 | 224 | th | 350 | 90 | 7 | K/Ar |
| 27-1 | L | g | CW | 51.67 | −120.04 | 327.8 | 71.1 | 0 | 8/10 | 202 | 3.9 | 70.7 | 175.4 | 6.2 | 81 | th | 350 | 90 | 7 | K/Ar |
| 27-2 | | g | CW | 51.67 | −120.04 | 314.2 | 66.0 | 10 | 10/10 | 101 | 4.8 | 60.8 | 162.9 | 7.3 | 45 | th | 350 | 90 | 7 | K/Ar |
| 27-3 | U | g | CW | 51.67 | −120.04 | 311.1 | 69.2 | 0 | 10/10 | 169 | 3.7 | 60.9 | 172.0 | 6.0 | 65 | th | 350 | 90 | 7 | K/Ar |
| 28-1 | | | CW | 51.63 | −120.13 | 332.1 | 69.3 | 9 | 9/10 | 200 | 3.6 | 73.0 | 165.5 | 6.0 | 74 | th | 350/500 | 90/50 | 6 | K/Ar |
| 29-1 | L | | KA | 49.83 | −119.75 | 3.3 | 81.9 | 0 | 11/11 | 355 | 2.4 | 65.6 | 242.3 | 4.7 | 96 | th | 760 | 110 | 8 | K/Ar |
| 29-2 | | h | KA | 49.83 | −119.75 | 345.9 | 77.6 | 0 | 10/11 | 821 | 1.7 | 72.0 | 221.8 | 3.1 | 241 | th | 760 | 110 | 8 | K/Ar |
| 29-3 | U | h | KA | 49.83 | −119.75 | 343.3 | 76.4 | 0 | 9/10 | 744 | 1.9 | 73.1 | 214.8 | 3.4 | 226 | th | 760 | 110 | 8 | K/Ar |

[a] Site identification number (ID) is composed of two numbers separated by a dash. Sites with the same number before the dash usually belong to stratigraphic sequences. Asterisk indicates a site that was not considered for paleomagnetic calculations. The number after the dash indicates the position of the lava flow in the sequence (upwards or downward). U/L indicate the uppermost and lowermost lava flows of these sequences. DG refers to directional group (see text for explanation). Volcanic field abbreviations are MM for Mount Meager, GL for Garibaldi Lake, ST for Silverthrone, CW for Clearwater - Wells Gray Park, and KA for Kelowna Area. D and I are the mean site declination and inclination; SC indicates the number of samples oriented with sun compass in the site; n/N are the number of samples used for calculation of site mean direction per number of processed samples. K is the dispersion parameter of directions (dir) or VGPs; $\alpha_{95}$ is the 95% confidence cone about the mean direction (Dir.) or mean VGP; AF/Th represent mean direction results after AF or thermal demagnetization. Age is the age(s) obtained in previous studies from the flow or volcanic unit (see text). U is the uncertainty range of the age. R is the reference source on which a given age is based: 1 is *Read* [1990] and *Hickson et al.* [1999], 2 is *Green et al.* [1988], 3 is *Green* [1977], 4 is *Roddick* [1996], 5 is *Hickson et al.* [1995], 6 is *Hickson and Souther* [1984], 7 is *Hickson* [1986], and 8 is *Church* [1980]. Meth. is the method used to obtain the age (Str., stratigraphic relationship).

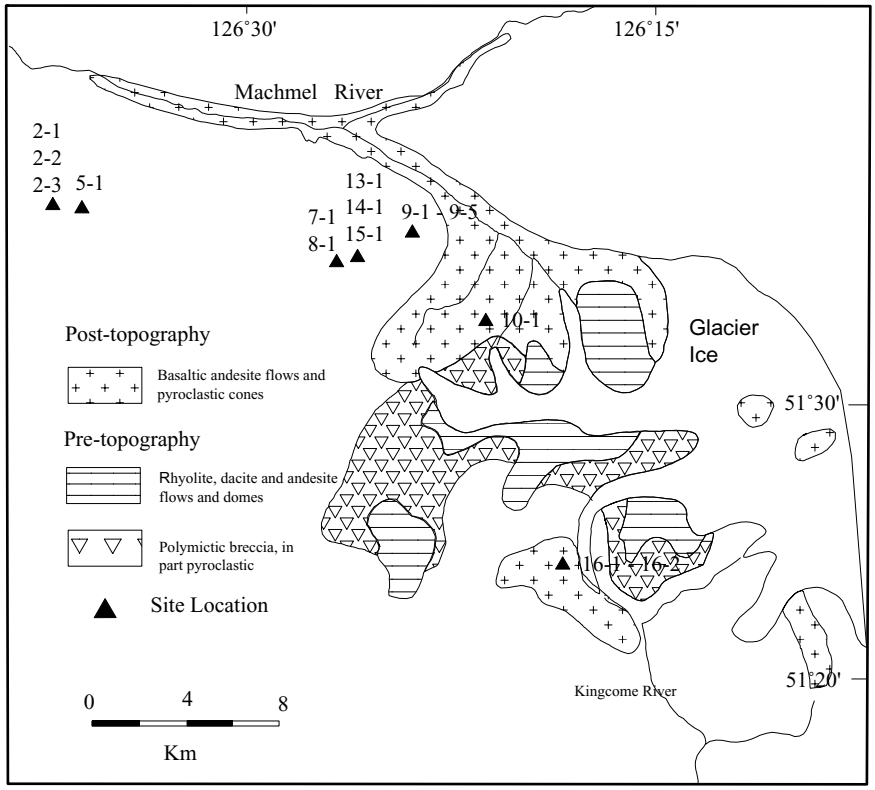



**Figure 2.** Geology of the Silverthrone volcanic field and sampling-site locations (triangles) (modified from *Green et al.* [1988]). The sites that fall outside the mapped area belong to the unit of basaltic andesite flows (crosses).

unit and underlie glacio-lacustrine sediments with a $^{14}C$ radiometric date of 34.2 Ka [*Green*, 1977]. Sites from this volcanic field are indicated in Table 1 by the abbreviation GL.

[9] The Mount Meager Volcanic Field has been active since the Pliocene [*Green et al.*, 1988] and has several volcanic centers. The only lava flow that was sampled here was a dacitic lava flow, erupted from Mount Meager along with other volcanic products (volcanic ash and pyroclastic flows) and has a $^{14}C$ radiometric date of 2350 yr BP [*Read*, 1990; *Hickson et al.*, 1999]. This lava flow was sampled twice (site 1-1 and site 1-2), and although the range of uncertainty of the directions at both sites overlap, the result of site 1-2 was discarded because of field observations and site 1-1 was preferred. Sites from this volcanic field are indicated in Table 1 by the abbreviation MM.

[10] Lava flows from the Clearwater-Wells Gray Park Volcanic Field (Figure 3) are exposed along the Clearwater and Murtle river basins. *Hickson and Souther* [1984] differentiated four volcanic assemblages that began forming 3 Ma ago: (1) an early glacial assemblage; (2) a deeply dissected, valley-filling assemblage composed of basaltic lava flows originating from different eruptive centers, and locally interstratified with sediments, which forms the main focus of sampling in this area; (3) a late intraglacial assemblage that includes ice-contact deposits and (4) post-glacial pyroclastic cones and lava flows. Twenty-eight basaltic lava flows (sites 20-1 to 28-1) were sampled from assemblage (2). We assigned ages to these flows, ranging between 0.2 and 0.5 Ma, based on K/Ar ages obtained from nearby outcrops which were studied previously (Table 1). Sites 18-1 and 19-1 were obtained from eroded lava conduits near Ray Mountain Volcano dated at 0.03 ± 0.03 Ma (K-Ar) [*Hickson et al.*, 1995]. Sites from this volcanic field are indicated in Table 1 by the abbreviation CW.

[11] Lavas from the Kelowna area (Table 1) were sampled from eroded Pleistocene volcanic outcrops. Three overlapping lava flows at Lambly

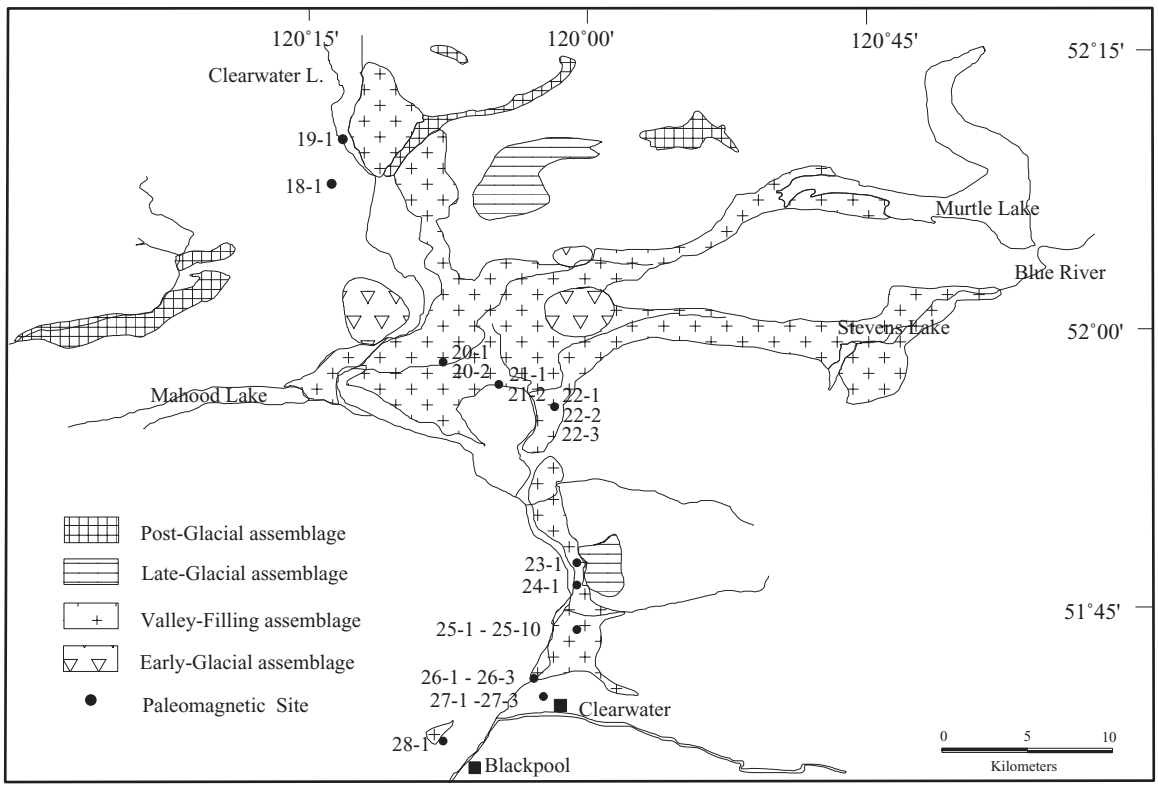



**Figure 3.** Geology of the Clearwater-Wells Gray Park volcanic field and sampling-site locations (dots) (modified from *Hickson and Souther* [1984]).

Creek, near Kelowna were sampled. A radiometric date of 0.76 ± 0.11 Ma (K/Ar) was given by *Church* [1980] to these flows. Sites from this volcanic field are indicated in Table 1 by the abbreviation KA.

## 3. Laboratory Analysis

[12] Most of the directions derived from sun and magnetic compass readings coincided within few degrees (<3°). Directions from sun compass readings were used at four sites (sites 1-1, 1-2, 1-5 and 26-3) in which some of the directions derived from sun compass and magnetic compass differed 5° or more.

[13] Stepwise AF demagnetization was performed on one sample per site. AF demagnetization was done using a Schonstedt AF demagnetizer and later a Dtech D-200 AF demagnetizer. All samples were thermally demagnetized in 14 to 21 steps utilizing a Schonstedt oven. The measurements were made in a 2G Cryogenic magnetometer in a shielded room at the University of Florida. Generally good agreement between AF and thermal demagnetization was observed (Figures 4a–4c) and the results from thermal demagnetization were used to calculate the primary component of magnetization of most sites. However, AF demagnetization gave better results (Figure 4d) at sites 23-1, 18-1, 19-1, and 24-1 and was used for analysis of the latter three sites. Magnetic susceptibility was monitored during thermal demagnetization, using a Sapphire susceptibility meter (except for samples from Clearwater-Wells Gray Park and Lambly Creek volcanic fields).

[14] The behavior of the basaltic rock material during laboratory analysis was in general typical for subaerial basalts. The NRM intensity varied between about 1 to 10 A/m, but was more commonly around 4 A/m. The principal component of magnetization was easily defined by removing the viscous remanent magnetization (VRM) in the first few demagnetization steps. In most cases a small (<5%) or no fraction of the NRM remained after the 570°C heating step, suggesting titanomagnetite as the magnetic carrier; however few sites still had

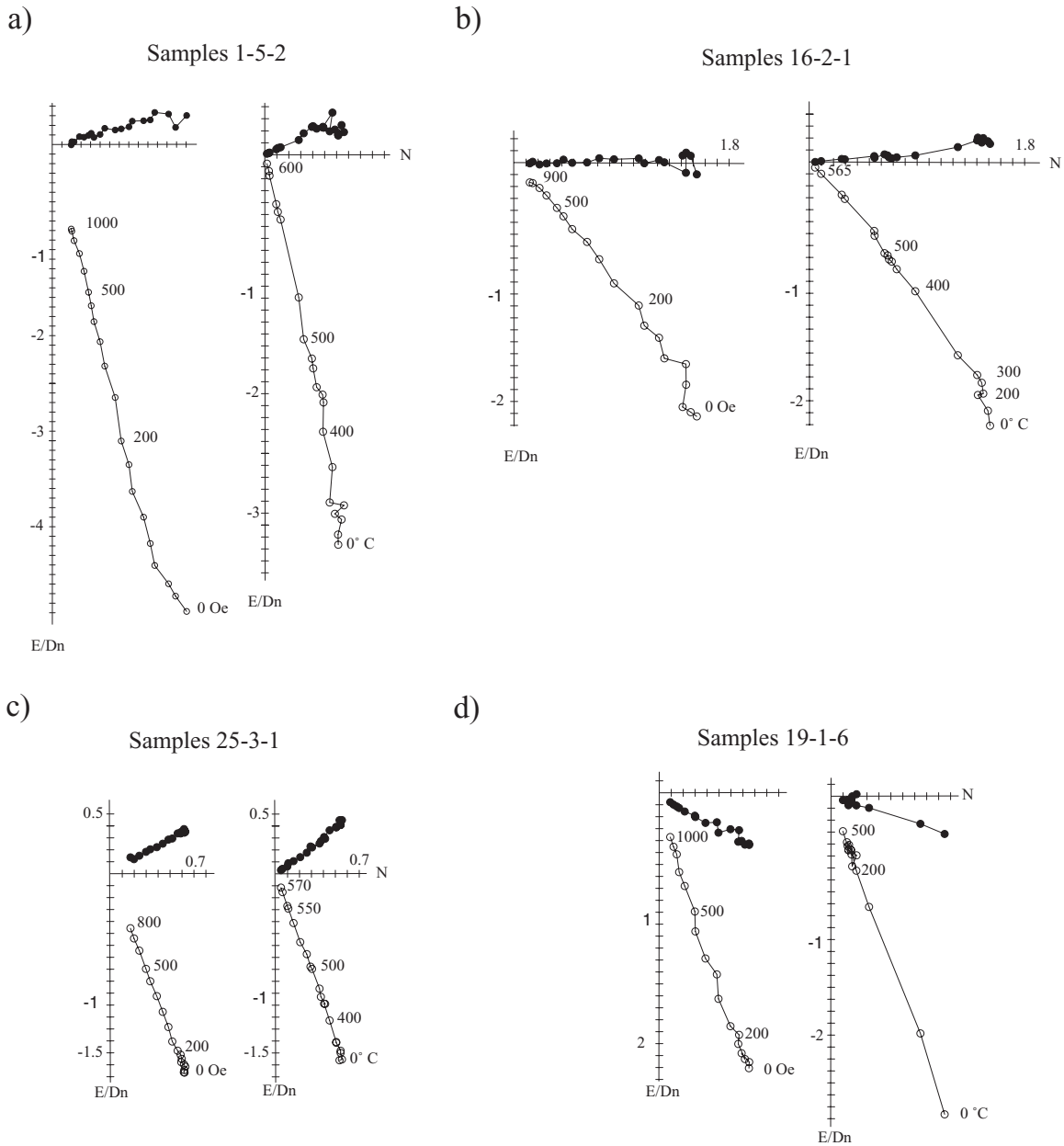



**Figure 4.** Zijderveld diagrams of AF and thermal demagnetization (left and right respectively) on replicate samples. Good agreement between the two methods is shown in Figures 4a–4c. Difficult to interpret thermal demagnetization curves relative to AF demagnetization, with both methods still indicating the same general direction is shown in Figure 4d. Approximate intensity values are in A/m.

>5% of the NRM remaining at 600°C (the highest temperature step applied) which can indicate the presence of small amounts of hematite.

## 4. Data Analysis and Selection Criteria

[15] In order to obtain suitable measurements of the paleomagnetic field and avoid spurious data that could blur real nondipole contributions to magnetic field models, it is necessary to establish stringent procedures and selection criteria for statistical parameters. The data analysis procedures and selection criteria used in this study closely follow those proposed by *Tauxe et al.* [2000] and supersede those of *McElhinny and McFadden* [1997]. The direction of the primary component

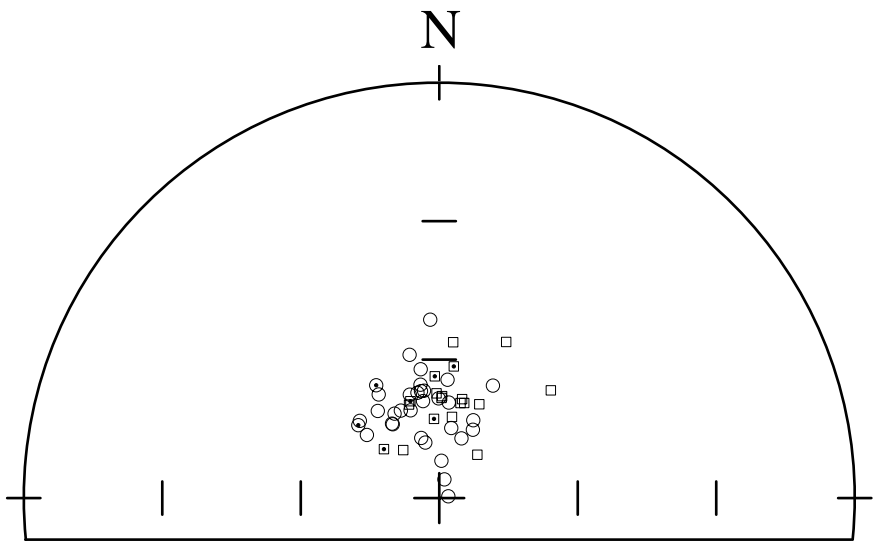



**Figure 5.** Equal area projection of site-mean directions. Squares represent directions of lava flows ≤150 Ka and circles represent directions of lava flows 200 to 760 Ka old. Rejected sites are shown with dots inside their respective symbols (squares or circles). All inclination point downward.

of magnetization was obtained through principal component analysis [*Kirschvink*, 1980] using a segment of at least five points of the demagnetization curve directed toward the origin with a maximum angular deviation (MAD) $\leq 5°$. Site-mean directions (Figure 5, Table 1) were obtained applying *Fisher* [1953] statistics from at least 5 cores and selected for further analysis if $\alpha_{95}$ values were $\leq 5°$ (rounded to the nearest integer). Based on these criteria, we selected 45 of the 52 flows sampled; that is seven sites were rejected (Figure 5).

[16] Secular variation analysis were made by calculating the angular standard deviation ($S_b$) of the virtual geomagnetic poles among the selected group of sites with respect to the Earth's axis of rotation using the method described by *Johnson and Constable* [1996] in which the effect of within site scatter is subtracted from the total scatter. The 95% confidence limits of the resultant scatter (angular standard deviation) were calculated using the method described by *Cox* [1969].

## 5. Results and Discussion

[17] The directions of sites within volcanic fields tend to cluster together indicating that their ages are closer in time compared to the sites from other volcanic fields. More precisely, it can be observed that younger sites (1-1 to 15-1) that are ≤150 Ka and older sites (16-1 to 29-3) that are 200 to 760 Ka (Table 1) show a distinct distribution of the paleomagnetic directions in time (Figure 5). The mean

**Table 2.** Statistical Data Among Groups of Sites[a]

| | D | I | N | K Dir | $\alpha_{95}$ Dir | VGP Long | VGP Lat | K VGP | $\alpha_{95}$ VGP | $S_t$ | $S_b$ | $S_u$ | $S_l$ | O.G | O.A |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Selected Sites | 356.9 | 70.2 | 45 | 57 | 2.8 | 215.8 | 85.5 | 23 | 4.5 | 17.7 | 17.5 | 20.5 | 15.2 | yes | yes |
| All Sites | 354.7 | 70.0 | 52 | 57 | 2.6 | 199.6 | 85.1 | 23 | 4.2 | 17.8 | 17.5 | 20.2 | 15.4 | yes | no |
| Filtered Sites | 0.0 | 69.4 | 33 | 55 | 3.4 | 237.3 | 87.2 | 23 | 5.4 | 16.9 | 16.7 | 20.0 | 14.3 | yes | yes |
| Older Flows | 349.5 | 70.7 | 31 | 65 | 3.2 | 191.0 | 82.5 | 25 | 5.3 | 18.0 | 17.8 | 21.6 | 15.2 | no | no |
| Younger Flows | 11.7 | 68.1 | 14 | 61 | 5.1 | 311.4 | 83.0 | 26 | 7.9 | 17.4 | 17.3 | 23.1 | 13.8 | yes | yes |

[a] Abbreviations for columns D to $\alpha_{95}$ VGP (first to ninth) are as in Table 1. Data of VGP scatter relative to the Earth's axis of rotation is given in columns: St (total scatter), Sb (corrected scatter), Su (upper 95% confidence limit of the scatter) and SL (lower 95% confidence limit of the scatter). O.G./O.A indicates whether the 95% confidence limits ($\alpha_{95}$) of the mean direction/mean VGP overlap the GAD/Earth's rotation axis.

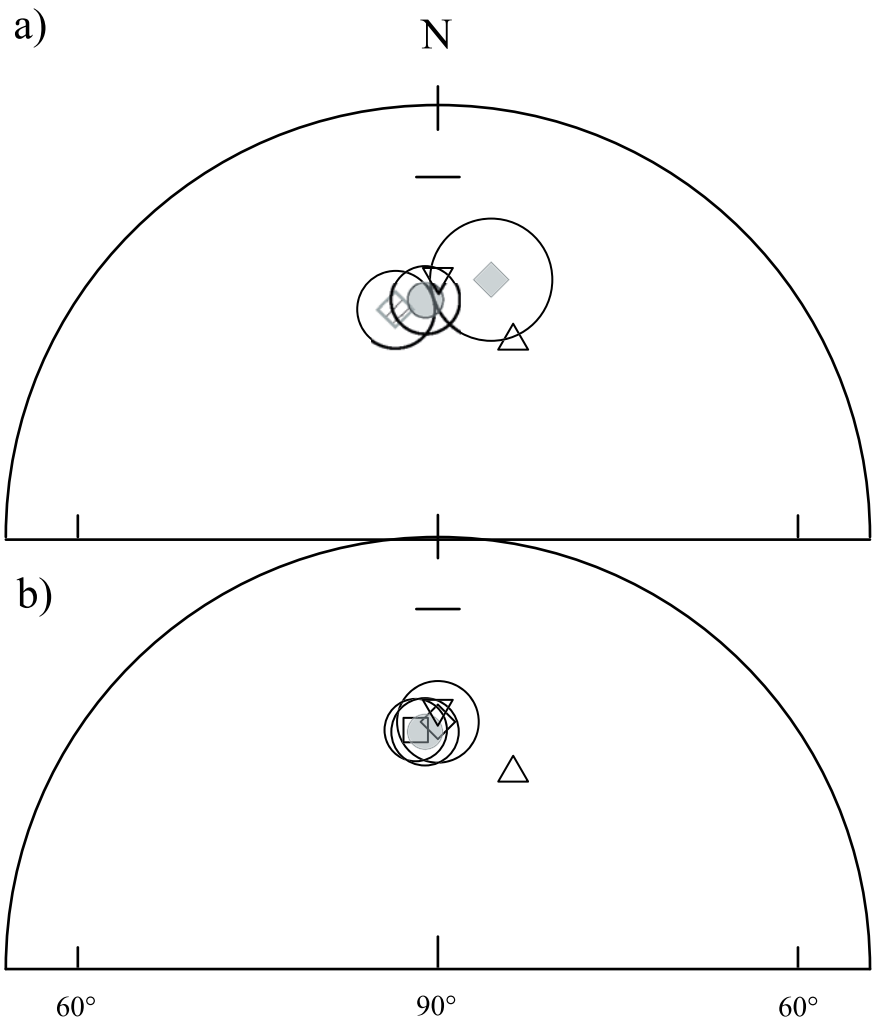



**Figure 6.** Equal area projection of mean directions from several groups of sites. Comparisons of the mean direction from the selected group of sites (gray-filled circle), GAD (triangle pointing down) and IGRF (triangle pointing up) are shown in Figure 6a with the mean direction among the younger group of lava flows (≤150 Ka) (gray-filled diamond) and the older group of lava flows (200 to 760 Ka) (striped diamond) and in Figure 6b with the mean direction of all lava flows (square) and filtered data (empty diamond).

directions among the younger and older lava flows have circles of confidence that barely overlap and only the mean direction from the group of younger lavas coincides with the GAD (Table 2, Figure 6). Given the age uncertainty of the lava flows and the episodic nature of volcanism, it is uncertain to what extent has the paleomagnetic field been sampled in time. That is, it is not known which fractions, corresponding to periods of enhanced volcanism, of the total period of time being studied (mainly the last 560 Ka) have been sampled.

[18] The mean direction of magnetization and the mean VGP of the 45 selected lava flows with $\alpha_{95}$ < 5° (Table 1) coincide at the 95% confidence level, with the expected direction of the GAD (Inc = 68.3°) in the area (mean latitude = 51.5°N) and the geographic north respectively (Figures 6 and 7). The directional distribution is fisherian while the VGP distribution is not. Similar results are observed when considering all the lava flows (including results from the seven lava flows that were rejected), but in this case, at the 95% confidence level, the mean direction almost falls away from the GAD and the mean VGP does not coincide with the Earth's axis of rotation. The shift in the direction obtained by taking into consideration the previously discarded sites produces a shift of the mean direc-

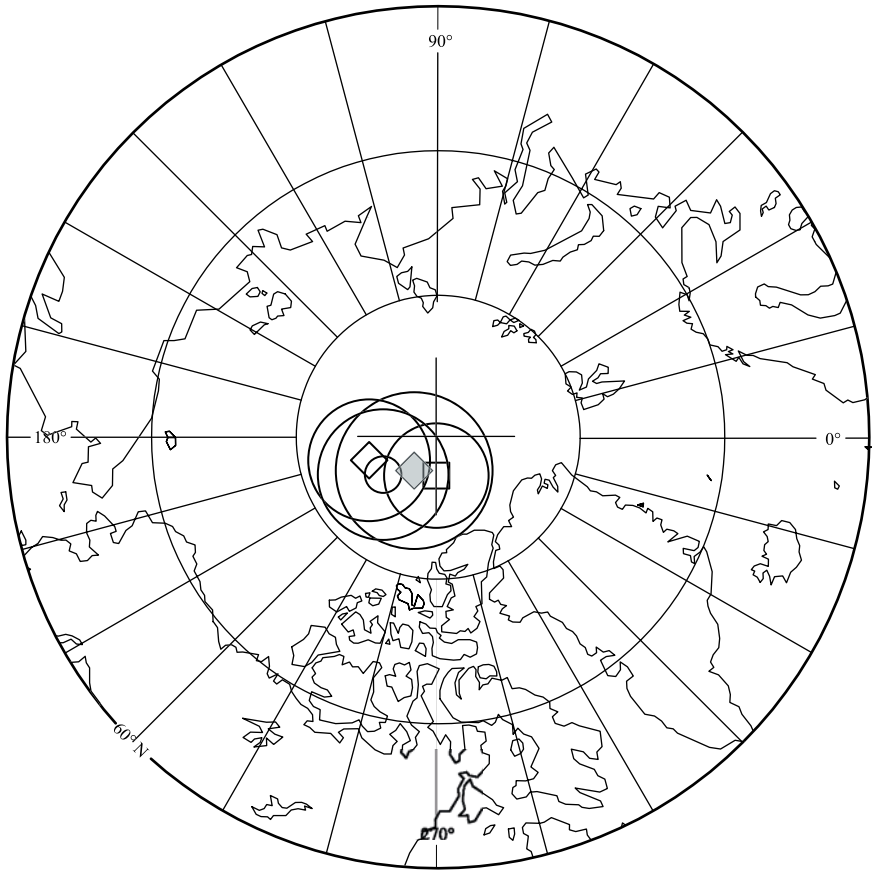



**Figure 7.** Mean virtual geomagnetic poles (VGPs) of selected sites (circle), all lava flows (empty diamond), filtered data (gray-filled diamond), and results [*Mitchell et al.,* 1989] from Indian Heaven volcanic field (square).

tion away from the GAD despite the fact that five of the rejected sites belong to the group of younger flows that tend to plot closer to the GAD. With a relatively small number of observations, it would be premature to argue that the application of selection criteria yields better results. Studies that explain the possible underlying reasons supporting the use of selection criteria are necessary.

[19] The mean direction of the selected sites lies somewhat closer to the group of older flows, because they are represented by more than twice as many flows as the group of younger lavas, although that does not imply an uneven sampling in time since the group of older flows represent about twice the period of time represented by the younger group (only the three flows from the Kelowna area are older than 560 Ka). This observation highlights the need for uniform sampling through time.

[20] Results of sites obtained from lava sequences deserve a closer examination because they are also likely to incorporate a bias by over-sampling a particular period of time. In order to investigate this problem, mean directions from contiguous lava flows having directions with overlapping circles of confidence were obtained, and used to calculate an overall mean direction along with the remaining flows that pass the selection criteria. *Mankinen et al.* [1985] applied this method and identified directional groups among the flows having overlapping circles of confidence. The same terminology is used here and flows conforming to directional groups are shown in Table 1. This procedure produces a mean direction and mean VGP (Table 2 and Figure 7) that is very similar to the unfiltered result, although $\alpha_{95}$ increases and the mean is slightly displaced toward the GAD.

[21] It is uncertain that the process of filtering the data benefits the quality of the results. The inclusion or exclusion of sites from the group of older or younger flows slightly shifts the balance between the two groups, driving the mean in one way or another. In this case more data points were taken out from the older group (13) than from the younger group (3), driving the mean (of filtered sites) toward the cluster of younger sites and toward the GAD (Figure 6). Instead, it is uncertain if contiguous lava flows with overlapping 95% circles of confidence record a relatively stationary direction of the field rather than that those flows erupted in a very short period of time [*Love*, 2000]. The validity of this concern is observed in the overlapping directions between flows 9-4 and 9-5 which are separated by a soil layer, thus indicating a substantial period of time between the two flows (for this reason these two flows were not included in the same directional group, see Table 1). Several cases in which filtering has been applied in lava sequences, have also produced mean directional results that do not differ substantially from those obtained from the unfiltered data [*McElhinny et al.*, 1996b; *Szeremeta et al.*, 1999; *Laj et al.*, 1999]. In general, it can be said that the relative low accumulation rate in the Garibaldi volcanic belt [*Sherrod and Smith*, 1990] and the presence of multiple volcanic centers in the Silverthrone and Wells-Grey Park volcanic fields helped to evenly sample the paleomagnetic field in time.

[22] The fact that the mean direction and mean VGP among the selected flows and filtered dataset coincide with the expected GAD and geographic pole, respectively, does not suggest a significant persistent contribution of the nondipole components of the field. Particularly the ''far-sided'' effect [*Wilson*, 1970], although expected to be relatively small at this latitude, that would cause the VGP to plot further from the North Pole with respect to the sampling area, is not observed. Likewise the small negative inclination anomaly ($-2°$ to $-4°$) for the area obtained as a result of modeling Brunhes-age paleomagnetic records from lava flows [*Johnson and Constable*, 1995] is not present in the dataset.

[23] The corresponding angular standard deviation of the VGP relative to the Earth's axis of rotation ($S_b$) for the selected sites is 17.5° with lower ($S_l$) and upper ($S_u$) 95% confidence limits [*Cox*, 1969] of 20.5° and 15.2° respectively (Table 2). These values are within the range expected from Model G [*McFadden et al.*, 1988] of secular variation for that latitude (17.4° with upper and lower confidence limits of 19.3° and 15.6° respectively). Similar scatter results are obtained when considering all the flows and after filtering (Table 2). The agreement of the VGP dispersion of this study with the dispersion value predicted in Model G suggests that the secular variation has been adequately sampled.

[24] The mean VGP reported from a similar study (in terms of age, rock type and quality) further south in the Indian Heaven Volcanic Field of Washington State [*Mitchell et al.*, 1989] plots as in this study, in a nearsided position although coinciding with the rotational axis at the 95% confidence level (Figure 7). The VGP scatter of 15.0° (with upper and lower limits of 17.9° and 13.2°) obtained in that study coincides less precisely than in this study with the scatter expected from Model G at 46° N (16.6° with lower and upper confidence limits of 15.0° and 18.3°).

## 6. Conclusions

[25] Paleomagnetic results from Brunhes age lava flows from a broad area of volcanic fields in Southern British Columbia are presented. Great effort has been placed in obtaining high quality results in order to obtain data sensitive to nondipole components of the field. The data set presented here does not support the unequivocal persistent presence of nondipole components of the field during the period of time studied. Mean directions and mean VGPs calculated from a selected group of sites and from filtered data coincide with the GAD and axis of rotation, respectively. However, the means calculated from all the sites studied do not. Filtering the data, with the objective of improving the quality of the results, produces a shift in the number of observations representing an older and a younger group of

lava flows which cluster in different directions, and that by itself alters the results.

[26] The mean VGP of these sites is similar to that obtained from the nearby Indian Heaven volcanic field, and the far-sided effect is not observed in these two studies. The VGP dispersion obtained in this study coincides with the one expected from model G and is higher that the one obtained in the Indian Heaven volcanic field.

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## References

Carlut, J., X. Quidelleur, V. Courtillot, and G. Boudon, Paleomagnetic directions and K/Ar dating of 0 to 1 Ma lava flows from La Guadalupe Island (French West Indies): Implications for time-averaged field models, *J. Geophys. Res.*, *105*, 835–849, 2000.

Church, B. N., Geological fieldwork, 1979: A summary of field activities, *Geol. Fieldwork*, *980–1*, 97–99, 1980.

Cox, A., Confidence limits for the precision parameter k, *Geophys. J. R. Astron. Soc.*, *18*, 545–549, 1969.

Fisher, R. A., Dispersion on a sphere, *Proc. R. Soc. London Ser. A*, *217*, 295–305, 1953.

Green, N. L., Multistage andesite genesis in the Garibaldi lake Area, southernmost British Columbia, Ph.D. thesis, Univ. of British Columbia, Vancouver, 1977.

Green, N. L., R. L. Armstrong, J. E. Harakal, J. G. Souther, and P. B. Read, Eruptive history and K-Ar geochronology of the late Cenozoic Garibaldi volcanic belt, southwestern British Columbia, *Geol. Soc. Am. Bull.*, *100*, 563–579, 1988.

Hickson, C. J., Quaternary volcanism in the Wells Gray-Clearwater area, East Central British Columbia, Ph.D. thesis, Univ. of British Columbia, Vancouver, 1986.

Hickson, C. J., and J. G. Souther, Late Cenozoic volcanic rocks of the Clearwater-Wells Gray area, British Columbia, *Can. J. Earth Sci.*, *21*, 267–277, 1984.

Hickson, C. J., J. G. Moore, L. Calk, and P. Metcalfe, Intraglacial volcanism in the Wells Gray - Clearwater Volcanic Field, east-central British Columbia, Canada, *Can. J. Earth Sci.*, *32*, 838–851, 1995.

Hickson, C. J., J. K. Russell, and M. V. Stasiuk, Volcanology of the 2350 B.P. eruption of Mount Meager volcanic complex, British Columbia, Canada: Implications for hazards from eruptions in topographically complex terrain, *Bull. Volcanol.*, *60*, 489–507, 1999.

Johnson, C. L., and C. G. Constable, The time-averaged geomagnetic field as recorded by lava flows over the past 5 Myr, *Geophys. J. Int.*, *122*, 489–519, 1995.

Johnson, C. L., and C. G. Constable, Paleosecular variations recorded by lava flows over the past five million years, *Phil. Trans. R. Soc. London Ser. A*, *354*, 89–141, 1996.

Johnson, C. L., and C. G. Constable, The time-averaged geomagnetic field: Global and regional biases for 0–5 Ma, *Geophys. J. Int.*, *131*, 643–666, 1997.

Johnson, C. L., J. R. Wijbrans, C. G. Constable, J. Gee, H. Staudigel, L. Tauxe, V. H. Forjaz, and M. Salgueiro, $^{40}Ar/^{39}Ar$ ages and paleomagnetism of Sao Miguel lavas, Azores, *Earth Planet. Sci. Lett.*, *160*, 637–649, 1998.

Kirschvink, J. L., The least squares line and plane and the analysis of palaeomagnetic data, *Geophys. J. R. Astron. Soc.*, *62*, 699–718, 1980.

Laj, C., H. Guillou, N. Szeremeta, and R. Coe, Geomagnetic paleosecular variation at Hawaii around 3 Ma from a sequence of 107 lava flows at Kaena Point (Oahu), *Earth Planet. Sci. Lett.*, *170*, 365–376, 1999.

Love, J. J., On the anisotropy of secular variation deduced from paleomagnetic volcanic data, *J. Geophys. Res.*, *105*, 5799–5816, 2000.

Mankinen, E. A., M. Prevot, C. S. Gromme, and R. S. Coe, The Steens Mountain (Oregon) geomagnetic polarity transition, 1, Directional history, duration of episodes, and rock magnetism, *J. Geophys. Res.*, *90*, 10,393–10,416, 1985.

McElhinny, M. W., and P. L. McFadden, Paleosecular variation over the past 5 Myr based on a new generalized database, *Geophys.J. Int.*, *131*, 240–252, 1997.

McElhinny, M. W., P. L. McFadden, and R. T. Merrill, The time-averaged paleomagnetic field 0–5 Ma, *J. Geophys. Res.*, *101*, 25,007–25,027, 1996a.

McElhinny, M. W., P. L. McFadden, and R. T. Merrill, The myth of the Pacific dipole window, *Earth Planet. Sci. Lett.*, *143*, 13–22, 1996b.

McFadden, P. L., R. T. Merrill, and M. W. McElhinny, Dipole/quadrupole family modeling of paleosecular variation, *J. Geophys. Res.*, *93*, 11,583–11,588, 1988.

Mitchell, R. J., D. J. Jeager, J. F. Diehl, and P. E. Hammond, Paleomagnetic results from the Indian Heaven volcanic field, south-central Washington, *Geophys. J. R. Astron. Soc.*, *97*, 381–390, 1989.

Read, P. B., Mount Meager Complex, Garibaldi Belt, southwestern British Columbia, *Geo. Can.*, *17*, 167–170, 1990.

Roddick, J. A., Geology, Rivers Inlet (92M)-Queens Sound (102P) British Columbia: Geological Survey of Canada, *Open File 3278*, scale 1:250,000, Natl. Resour. Canada, Ottawa, Ontario, 1996.

Sherrod, D. R., and J. G. Smith, Quaternary extrusion rates of the Cascade Range, northwestern United States and southern British Columbia, *J. Geophys. Res.*, *95*, 19,465–19,474, 1990.

Szeremeta, N., C. Laj, H. Guillou, C. Kissel, A. Mazaud, and J. C. Carracedo, Geomagnetic paleosecular variation

in the Brunhes period, from the island of El Hierro (Canary Islands), *Earth Planet. Sci. Lett.*, *165*, 241–253, 1999.

Tauxe, L., H. Staudigel, and J. R. Wijbrans, Paleomagnetism and $^{40}Ar/^{39}Ar$ ages from La Palma in the Canary Islands, *Geochem. Geophys. Geosyst.*, *1*, Paper number 2000GC000063, 2000.

Wilson, R. L., Permanent aspects of the earth's nondipole magnetic field over Upper Tertiary times, *Geophys. J. R. Astron. Soc.*, *19*, 417–437, 1970.